# Supplementary Material: Grey-Box Online Identification for Pushrod-Driven Dexterous Hands

Supplementary Material

## Tables from Main Paper

For reader convenience, we reproduce the key tables from the main paper here.

TABLE I
ALGORITHM COMPARISON: REPRESENTATIVE $R^2$ VALUES (JOINTS 3–6)

| **Algorithm** | **Joint 3** | **Joint 4** | **Joint 5** | **Joint 6** |
|---|---|---|---|---|
| URDF-only (baseline) | -8.88 | -9.60 | -2.28 | -3.69 |
| Batch LS Hybrid | 0.03 | 0.03 | 0.17 | 0.26 |
| Grey-Box Standard NLMS (5-param) | **0.97** | **0.97** | **0.96** | **0.95** |
| FFLS ($\lambda = 0.95$) | $\approx$ -1000 | $\approx$ -1000 | $\approx$ -1000 | $\approx$ -1000 |
| UD-RLS (Bierman) | $\ll 0$ | $\ll 0$ | $\ll 0$ | $\ll 0$ |
| QR-RLS | $\ll 0$ | $\ll 0$ | $\ll 0$ | $\ll 0$ |
| EKF Parameter Est. | $\ll 0$ | $\ll 0$ | $\ll 0$ | $\ll 0$ |
| Neural Networks (best) | -10.50 | -8.32 | -12.15 | -9.87 |

TABLE II
GREY-BOX STANDARD NLMS RESULTS: $R^2$ AND RMSE FOR ALL SIX JOINTS

| **Joint** | **Name** | $\mathbf{R^2}$ | **RMSE (N·m)** |
|---|---|---|---|
| 1 | Thumb flex. | 0.9549 | 0.016031 |
| 2 | Thumb yaw | 0.8946 | 0.015653 |
| 3 | Index | 0.9723 | 0.000882 |
| 4 | Middle | 0.9700 | 0.000739 |
| 5 | Ring | 0.9582 | 0.002043 |
| 6 | Pinky | 0.9511 | 0.001188 |

TABLE III
PARAMETER CONTRIBUTION TO TOTAL TORQUE (PERCENTAGE, REPRESENTATIVE FINGER DATASET)

| **Parameter** | **Joint 3** | **Joint 4** | **Joint 5** | **Joint 6** |
|---|---|---|---|---|
| $C_f$ (Coulomb friction) | 18.2% | 15.3% | 19.1% | 17.8% |
| $C_v$ (Viscous friction) | 8.5% | 7.2% | 9.3% | 8.1% |
| $K_{q,\mathrm{L}}$ | 12.4% | 11.2% | 13.5% | 12.8% |
| $K_{\mathrm{c}}$ ($\dot{q}_i \cdot q_i$) | **46.5%** | **53.9%** | **48.2%** | **49.1%** |
| $O$ | 5.5% | 3.3% | 2.7% | 3.7% |
| **Sign of** $K_{\mathrm{c}}$ | – | – | – | – |

## Algorithm Benchmarking Details

This supplementary document provides the detailed failure-mode analysis for all 37 evaluated algorithms that is only summarized in the main paper.

### A. Algorithm Families and Typical Failure Modes

Beyond the representative $R^2$ values reported in Table I, the 37 evaluated algorithms exhibit distinct failure modes:

- **RLS/EKF variants (10 algorithms).** Standard RLS, FFLS, UD-RLS, QR-RLS and EKF parameter estimation consistently diverge or saturate ($R^2 \ll 0$, FFLS $\approx -10^3$). Covariance safeguards (eigenvalue clipping, resets, gain limiting) trigger in $> 80\%$ of update steps, indicating numerical breakdown of $(\Phi_t^\top \Phi_t)^{-1}$ under extreme ill-conditioning ($\kappa(\Phi_t^\top \Phi_t) > 10^7$).
- **Classical LS/LMS family (10 algorithms).** Batch LS remains numerically stable but limited to offline use and low predictive accuracy ($R^2 = 0.03$–$0.26$ on joints 3–6). Standard LMS and plain NLMS variants are sensitive to feature-scale imbalance when used without the grey-box residual structure and physical projection constraints. In contrast, Grey-Box Standard NLMS remains both stable and accurate ($R^2 \approx 0.96$ on joints 3–6 using optimized step sizes).
- **Sampling-based and nonparametric methods.** Sampling-based and nonparametric variants were less suitable for embedded online identification because they either required substantially higher per-update computation or depended on growing memory with the number of samples. This makes them unattractive for sustained 100–200 Hz operation on the target hand.
- **Black-box neural baselines.** Neural network variants exhibited overfitting in the low-SNR residual-dynamics setting and did not match the held-out accuracy of the constrained grey-box estimator. The dominant failure mode is memorization of high-frequency noise in the residual dynamics, exacerbated by the absence of explicit hard physical constraints.

### B. Ill-Conditioning and EKF Instability

RLS/EKF failure is attributed to extreme ill-conditioning: condition numbers exceed $10^7$–$10^9$, amplifying matrix inversion errors by $10^7$–$10^9$. For $\kappa = 10^8$, numerical errors ($\sim 10^{-6}$) are amplified to $> 10^2$, causing catastrophic failure. Even with comprehensive safeguards, safeguards trigger in $> 80\%$ of steps, causing the algorithm to lose learning capability.

*a) QR-RLS and UD-RLS Failure Analysis.:* QR-RLS (QR decomposition-based RLS) and UD-RLS (Bierman's UD factorization) are designed to avoid explicit matrix inversion by maintaining factorized representations of the covariance matrix, theoretically providing better numerical stability than standard RLS. However, in our extreme ill-conditioning regime ($\kappa(\Phi_t^\top \Phi_t) > 10^7$), both algorithms fail consistently ($R^2 \ll 0$). The fundamental issue is that factorization-based methods still operate on the same ill-conditioned matrix $\Phi_t^\top \Phi_t$, and the condition number propagates through the factorization process. For QR-RLS, the condition number of the R factor equals that of the original matrix, and small errors in the Q factor accumulate over time. For UD-RLS, the U and D factors become increasingly ill-conditioned as the algorithm runs, with diagonal elements spanning orders of magnitude ($D_{ii}$ ranging from $10^{-9}$ to $10^2$), causing loss of precision in floating-point arithmetic. When safeguards (eigenvalue clipping, resets, gain limiting) trigger in $> 80\%$ of update steps, the algorithms effectively stop learning, as parameter updates are constantly reset or clipped. In contrast, Standard NLMS avoids this issue entirely by normalizing the update step size using the scalar $\|\phi_i(t)\|^2$, which bounds the effective condition number of the update to approximately 1.2–1.3, independent of the condition number of $\Phi_t^\top \Phi_t$.

*b) EKF Overfitting and Instability.:* In our experiments, EKF-based parameter estimation either overfits or diverges numerically: it can drive the full-dataset $R^2$ close to 1, but on held-out data and on the final preprocessed datasets it yields strongly negative $R^2$ or even blow-ups. We therefore regard EKF as impractical for this extreme low-inertia, high-friction setting and focus on Grey-Box Standard NLMS as the stable and accurate online method.

### C. Why Grey-Box Standard NLMS Succeeds

Grey-Box Standard NLMS avoids matrix inversion via scalar normalization $\|\phi_i(t)\|^2$, maintaining well-conditioned updates ($\kappa \approx$ 1.2–1.3) even when $\Phi^T\Phi$ is severely ill-conditioned ($\kappa > 10^7$). The normalized step size $\mu/\|\phi_i(t)\|^2$ provides sufficient numerical stability, while the grey-box residual formulation and physical projection constraints keep the learned parameters interpretable. The resulting update has $O(n_p)$ complexity, enabling real-time implementation at 100–200 Hz.

### Neural Network Baselines and Alternative Online Methods

Neural network baselines were used as diagnostic comparisons for the residual-dynamics prediction task. Despite feature normalization and regularization, these black-box models were prone to overfitting and did not match the held-out accuracy of Grey-Box Standard NLMS on the final preprocessed dataset. Failure is attributed to: (1) low signal-to-noise ratio in the residual torque, (2) limited independent excitation data relative to model flexibility, (3) lack of explicit hard physical constraints, and (4) computational overhead compared with the lightweight NLMS update. Grey-Box Standard NLMS succeeds because it combines structural decomposition, scalar adaptive normalization, and projection onto physically meaningful parameter bounds.

**Comparison with other online methods:** Beyond RLS/EKF variants and neural networks, we evaluated adaptive-filtering, sampling-based, and nonparametric alternatives as diagnostic baselines. Methods that maintain matrix inverse estimates inherit the ill-conditioning problems discussed above, while sampling-based and nonparametric methods impose higher compute or memory costs than are desirable for sustained embedded operation. These comparisons highlight the practical value of the grey-box NLMS combination: numerical stability, low per-update cost, and high held-out accuracy under the evaluated excitation protocol.

### URDF Sensitivity and Cross-Dataset Robustness

Sensitivity analysis shows the grey-box approach is robust to moderate URDF calibration errors: $R^2$ degrades by $< 2\%$ for $\pm$5–10% inertia perturbations and 5–8% for $\pm 15\%$ errors, with $K_{c,i}$ preserving its sign pattern ($K_{c,1} > 0$ and $K_{c,i} < 0$ for $i = 2, \dots, 6$). Three complementary analyses validate that $K_{c,i}$ is not compensating for URDF errors: (1) *Orthogonality*: In periodic trajectories, $\ddot{q}_i$ and $\dot{q}_i$ are nearly orthogonal, so acceleration errors $\Delta I_{U,i} \cdot \ddot{q}_i$ project weakly into the velocity-position regression space. (2) *Correlation*: Correlation coefficient $|\rho| < 0.15$ between $I_{U,i} \cdot \ddot{q}_i$ and $\dot{q}_i \cdot q_i$ across all joints. (3) *Stability*: When $I_{U,i}$ varies by $\pm 10\%$, $K_{c,i}$ changes by only 5–8%, while a pure compensation term would change linearly. These results support the interpretation of $K_{c,i}$ as an independent physical mechanism rather than a URDF error compensator.

We also assess generalization across four independently collected sessions/datasets. Models maintain consistent performance: $R^2$ varies by $< 5\%$ for finger flexion joints and $< 10\%$ for thumb joints, with inter-dataset variance significantly smaller than intra-dataset variance ($p < 0.01$, F-test). Parameter estimates show similar consistency: $K_{c,i}$ values vary by $< 15\%$ across datasets, with CV $< 20\%$ for $K_{c,i}$ compared to $> 40\%$ for some other parameters, suggesting the coupling term is a fundamental physical property rather than a fitting artifact.

### Online Adaptation to Time-Varying Parameters

The Grey-Box Standard NLMS framework's rapid convergence (30–50 seconds) enables real-time adaptation to slow parameter drift (temperature changes, lubrication degradation, mechanical wear). Parameters converge to steady-state values (CV $< 5\%$ after 10,000–15,000 samples), which are reported in Tables II and III. During continuous operation, friction coefficients adapt as internal heating reduces lubricant viscosity, with estimates stabilizing within 10,000–15,000 samples (2–3 minutes) of any parameter change. This online capability distinguishes our approach from offline batch methods.

### Experimental Data

The experimental data and URDF models are available in the public repository at https://github.com/ZhongYijun/iros_supplementary. The datasets are organized in the `experimental_data` and `urdf_models` folders and include joint position, velocity, acceleration, motor current measurements, and current-derived equivalent joint effort/torque estimates collected at 100 Hz as described in the main paper (Section 5.1).